

The AI Apocalypse Matrix

BY ELIZA STRICKLAND & GLENN ZORPETTE

WHAT SHOULD WE MAKE OF OpenAI's GPT-4, anyway? Is the large language model a major step on the way to an artificial general intelligence (AGI)—the insider's term for an AI system with a flexible human-level intellect? And if we do create an AGI, might it be so different from human intelligence that it doesn't see the point of keeping *Homo sapiens* around?

If you query the world's best minds on basic questions like these, you won't get anything like a consensus. Some AI specialists, including the revered researcher Geoffrey Hinton, view GPT-4 as a clear sign of the imminence of AGI. Others, such as Rodney Brooks, assert that we're no closer to an AGI than we were 30 years ago.

On the topic of the potential of GPT-4 and its successors to wreak civilizational havoc, there's similar disunity. One of the earliest doomsayers was Nick Bostrom; long before GPT-4, he argued that once an AGI far exceeds our capabilities, it will find ways to escape the digital world and methodically destroy human civilization. On the other end are people like Yann LeCun, who rejects such scenarios as sci-fi twaddle.

In between are researchers who worry about the abilities of GPT-4 and future instances of generative AI to cause major disruptions in employment, to exacerbate the biases in today's society, and to generate propaganda, misinformation, and deep fakery on a massive scale. Worrisome? Yes, extremely so. Apocalyptic? No.

Many worried AI experts signed an open letter in March asking all AI labs to immediately pause "giant AI experiments" for six months. Then, at the end of May, an overlapping set of experts—academics and executives—signed a one-sentence statement urging the world to take seriously the risk of "extinction from AI."

Here, *IEEE Spectrum* has distilled the published pronouncements of 23 AI luminaries on large language models, the likelihood of an AGI, and the risk of civilizational havoc. We then used our best judgment to place them on this matrix. ■

More details on the positions of the AI experts listed here can be found in the online version of this article, "The AI Apocalypse: A Scorecard"



THE 23 AI SPECIALISTS FEATURED ARE...



SAM ALTMAN
CEO, OpenAI



MARC ANDREESSEN
Cofounder, Andreessen Horowitz



JACOB ANDREAS
Professor, MIT



EMILY M. BENDER
Professor, University of Washington



YOSHUA BENGIO
Professor, University of Montreal



NICK BOSTROM
Professor, University of Oxford



RODNEY BROOKS
Professor emeritus, MIT, and cofounder, RoboAI



SÉBASTIEN BUBECK
Research manager, Microsoft Research



JOY BUOLAMWINI
Founder, Algorithmic Justice League



TIMNIT GEBRU
Founder, Distributed AI Research Institute



ALISON GOPNIK
Professor, UC Berkeley



DAN HENDRYCKS
Director and cofounder, Center for AI Safety



GEOFFREY HINTON
Professor, University of Toronto



CHRISTOF KOCH
Chief scientist of the MindScope program, Allen Institute



JARON LANIER
Computer scientist, entrepreneur, author, artist



YANN LECUN
Chief AI scientist, Meta



GARY MARCUS
Professor, NYU



MARGARET MITCHELL
Chief ethics scientist, Hugging Face



MELANIE MITCHELL
Professor, Sante Fe Institute



ANDREW NG
Founder and CEO, Landing AI



MAX TEGMARK
Professor, MIT and president, Future of Life Institute



MEREDITH WHITTAKER
President, Signal Foundation



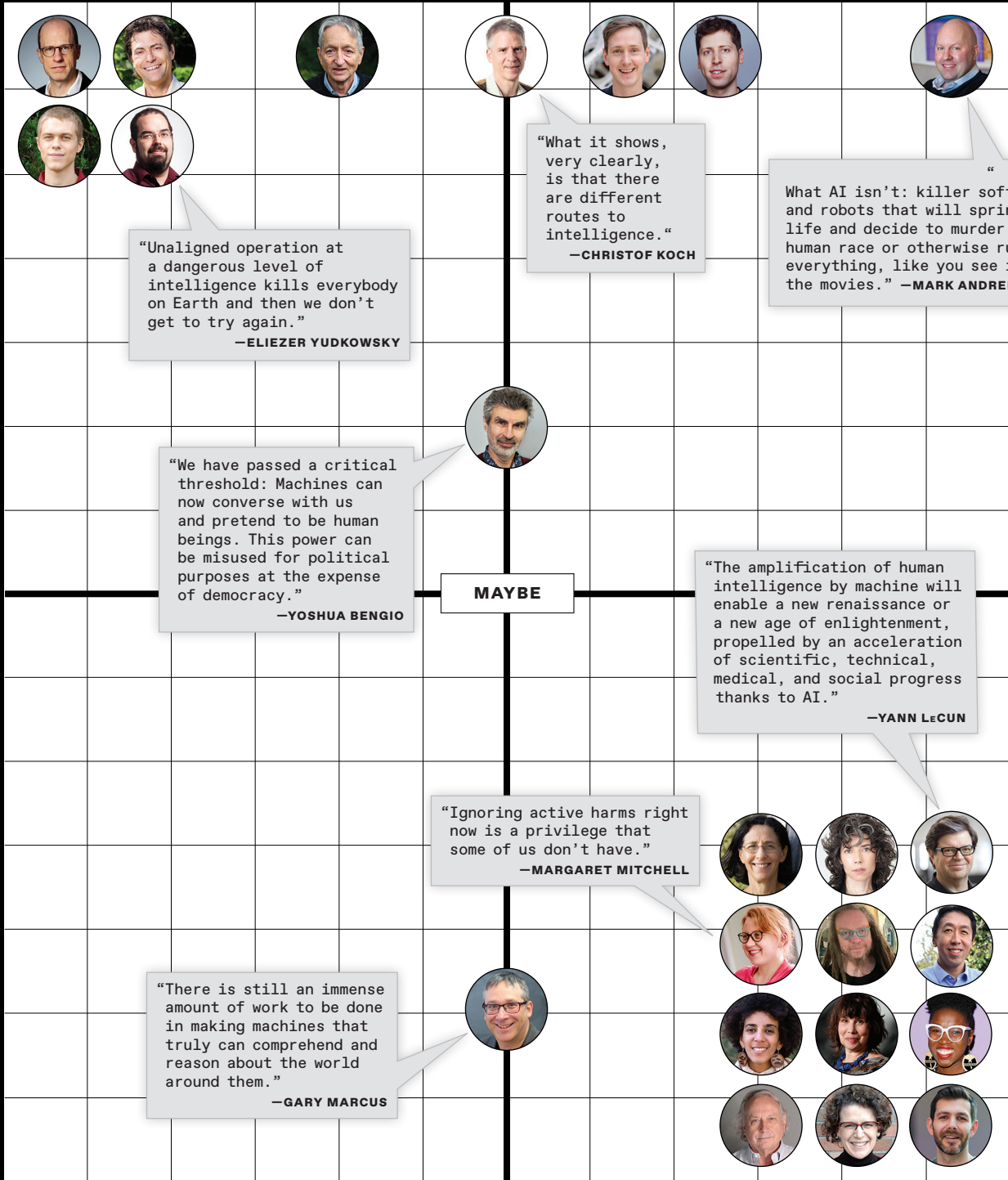
ELIEZER YUDKOWSKY
Cofounder, Machine Intelligence Research Institute

CLOCKWISE FROM TOP RIGHT: ANDREESSEN HOROWITZ; EMILY M. BENDER; UNIVERSITY OF OXFORD; MICROSOFT RESEARCH; TIMNIT GEBRU; DAN HENDRYCKS; ALLEN INSTITUTE; META; HUGGING FACE; ANDREW NG; SIGNAL FOUNDATION; MACHINE INTELLIGENCE RESEARCH INSTITUTE; MAX TEGMARK; MELANIE MITCHELL; NYU; JARON LANIER; UNIVERSITY OF TORONTO; GARY DOAK/ALAMY; GETTY IMAGES; CHRISTOPHER MICHEL; CAMILLE GLADU-DROUIN; GRETCHEN ERTL/MIT; OPENAI

HIGH PROBABILITY OF AGI

HIGH PROBABILITY OF DISASTER BY AGI

LOW PROBABILITY OF DISASTER BY AGI



LOW PROBABILITY OF AGI